THE AI TOOLBOX · CLASS 06

The Research Toolbox

Assemble the toolkit that turns a **curious** question into a **grounded** one.

Merit Point Academy · 60 min · press **S** notes · → next · 💬 to comment

RECAP · WHERE WE LEFT OFF

Last class in 60 seconds

Everything is a table

Rows = examples, columns = variables. See the DataFrame and the code gets simple.

Four verbs

Load → **inspect** → **select** → **summarize** — 80% of data work, and every question is a filter.

Honest holes

Find missing values, drop or fill — and **disclose** what you did.

Homework share: show your first Colab chart — and one debugging win (the error + what your AI co-pilot taught you). Today we zoom back out from code to **strategy**: the tools that find the right papers and datasets so your pandas skills point at something worth analyzing.

AGENDA · HOW THIS HOUR RUNS

Where we're going

- **0–8'** Homework share + a cure hidden between two libraries
- **8–20'** The modern research stack · reading citation graphs
- **20–32'** Four discovery tools, one each: Scholar · Elicit · Semantic Scholar · Connected Papers
- **32–40'** The 5-step literature workflow · GitHub for research
- **40–48'** Live demos: Elicit · Semantic Scholar · Connected Papers
- **48–54'** Two labs: explore a citation graph · rank papers by relevance
- **54–60'** Your literature map, homework & wrap-up
- Quizzes & 💬 comments throughout

By the end: use AI search tools to find the papers that matter fast, read a citation graph, and build a small literature map for your own question.

WARM-UP · ACTIVATE

Quick check · tap an answer

There are 10,000 papers on pedestrian detection. How does a researcher find the handful that actually matter — without reading all 10,000?

- Read the first 10 Google results and hope
- Use tools that search by meaning, summarize findings, and map citations — then read the few that survive
- Read only the most recent paper; it contains everything
- Ask an LLM and cite whatever it lists

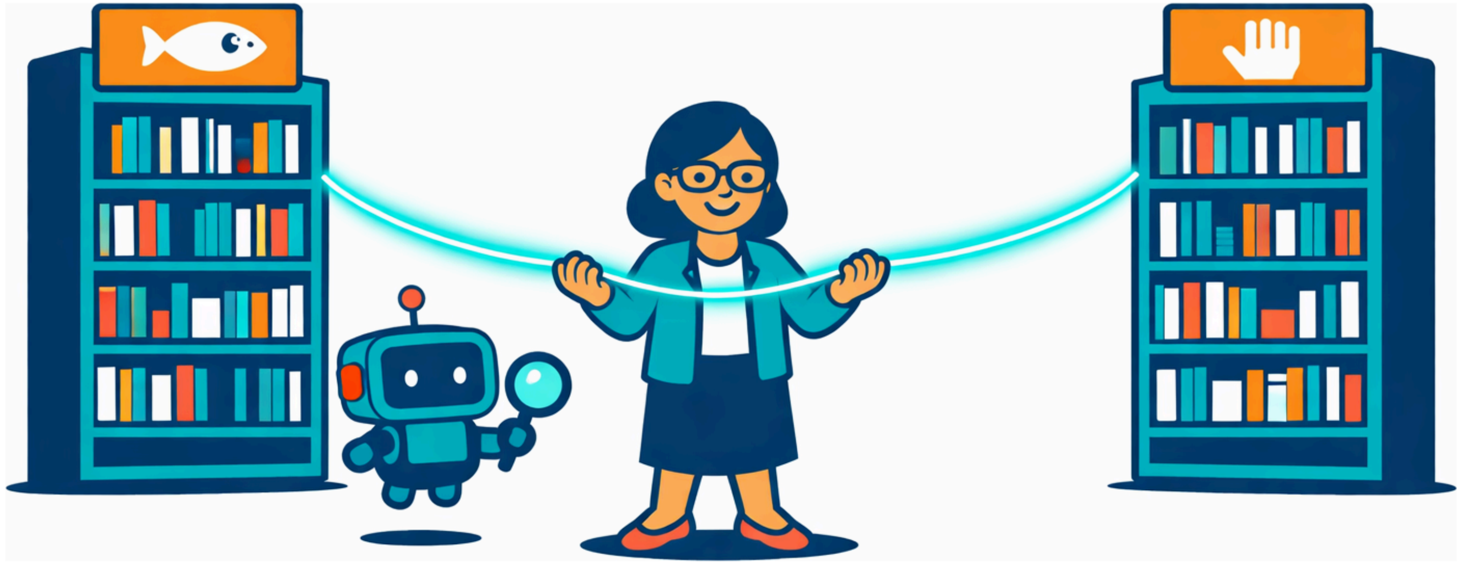
THE STORY

A cure hidden between two libraries

In 1986, librarian **Don Swanson** noticed something strange. One set of medical papers said **fish oil thins blood and improves circulation**. A completely separate set said patients with **Raynaud's disease suffer from poor circulation**. Nobody had connected them. Swanson did — purely by **reading the literature** — and proposed fish oil might help Raynaud's. Later clinical trials showed **he was right**.

🔊 Speak this story

Neural voice · click to play / pause



No lab. No experiment. The discovery was already sitting in the library — in the **gap between** two fields that never cited each other. Today's tools are built to find exactly those gaps.

CONCEPT 1 · YOU NO LONGER START FROM A BLANK BOX

The modern research stack

Four specialized tools, each best at a different job in the discovery loop:

🔎 Google Scholar

The broad index. Almost every paper ever written, with citation counts and "cited by" links. Your widest net.

💬 Elicit

Ask a question, get papers + findings. Type a research question in plain English; it returns relevant papers with their results already summarized in a table.

🧠 Semantic Scholar

Search by meaning, not keywords. AI TL;DR summaries, influence scores, and semantically related papers — from the Allen Institute for AI.

🕸 Connected Papers

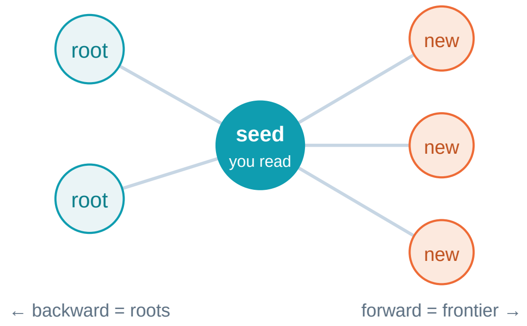
A visual citation graph. Paste one paper; see the whole neighborhood of related work as a map you can explore.

KEY IDEA — use the tools to **find** papers; then **read the papers yourself**. The tools narrow the field and summarize — but Class 4's rule still holds: never cite what you haven't opened.

CONCEPT 2 · THE IDEA IN ONE PICTURE

Citation graphs: a map of a whole field

- **Backward citations = the roots.** The papers your paper cites — the foundations it stands on.
- **Forward citations = the frontier.** The newer papers that cite yours — where the field went next.
- **Clusters = sub-fields.** Tight knots of papers citing each other reveal a specialty.



Takeaway: follow citations **back** to the roots of an idea and **forward** to what's new. Swanson's fish-oil discovery lived in a missing edge between two clusters — the graph makes those gaps visible.

CHECK · READING THE GRAPH

Quick check · tap an answer

You've found the one seminal paper for your topic. To discover the newest work building on it, which direction do you follow?

- Backward citations — the papers it cites
- Forward citations — the papers that cite it
- Neither; recency can't be found from citations
- Only papers by the same author

Google Scholar — the widest net

What it's best at

- Broadest coverage — nearly every paper, field-agnostic
- **Citation counts** = a quick impact signal
- **"Cited by"** and **"Related articles"** links = forward citations, one click away

Power moves

- Quotes for exact phrases; `author:` and year filters to narrow
- Sort by date for the frontier; by citations for the classics
- Set an alert to auto-track new papers on your topic

> You've used it since Class 2 (verifying LLM citations) and Class 4 (the 2-minute check). Today it becomes your **starting** tool, not just your fact-checker — the widest net you cast first.

Elicit — ask a question, get a findings table

No keyword guessing — type your **actual research question**. Elicit finds relevant papers and pulls out each one's key findings into a comparison table.

```
You ▸ Does adding night-time images improve pedestrian detection at night?
```

Returns: a table of papers × columns like *method* · *dataset* · *main finding* · *limitation* — hours of skimming, compressed to minutes.

Watch for

It's AI-powered, so the same rule applies: the summaries are **leads to verify**, not gospel. Open the paper before you trust the row. But as a **first pass** over a new field, nothing is faster.

The tabular superpower: that findings table is a ready-made comparison — it turns "read 20 papers" into "scan one spreadsheet, then read the 5 that matter." Your Class-4 summary framework, automated across a whole shelf.

Semantic Scholar — search by meaning

🧠 Semantic search

Understands the **meaning** of your query, not just keyword matches — finds papers that use different words for the same idea.

📝 TL;DR summaries

An AI one-sentence summary on each paper — triage a search page in seconds before opening anything.

📈 Influence score

Ranks papers by citation-network impact, so the field-defining work rises above the noise.

Built by the **Allen Institute for AI**. Where Scholar casts the widest net, Semantic Scholar casts the **smartest** one — great when you don't know the exact keywords a field uses (which is most of the time in a new field).

THE STACK · TOOL 4 OF 4

Connected Papers — see the neighborhood

How it works

Paste **one** paper you like. It builds a **graph** of the most related papers — connected by shared references and citations, not just direct links. Similar papers sit close; the field's structure appears.

Three views

- **Graph:** the whole neighborhood — spot clusters and bridges
- **Tree:** foundations → derivatives, the roots-to-frontier flow
- **List:** sorted by relevance for quick reading

This is concept 2, made real. One seed paper → a citation graph you can click through. The fastest way to grasp an unfamiliar field's shape — and to spot the **Swanson gaps** where two clusters almost touch.

CONCEPT 3 · THE META-SKILL

No best tool — only the right tool for the step

Broad first sweep? → Google Scholar or Perplexity

New field, don't know the words? → Semantic Scholar or Elicit

Ask a question, want findings? → Elicit's tables

See how a field connects? → Connected Papers

Quick summary of a topic? → Perplexity or ChatGPT (then verify)

Share code & results? → GitHub

> **The real skill isn't any one tool — it's chaining them.** A first sweep feeds a semantic search feeds a citation map feeds your reading list. Tools are a **relay team**, not rivals. *"Tools serve you — don't become a slave to tools."*

THE RELAY, IN ORDER · YOUR LITERATURE WORKFLOW

From blank page to reading list, in five steps

1. **Overview**

   Perplexity: "latest advances in ___?" — get the lay of the land

2. **Core papers**

   Scholar / Semantic Scholar: highly-cited + recent — pick your seed

3. **Map**

   Connected Papers: paste the seed → see the neighborhood

4. **Summarize**

   Elicit / ChatGPT: findings, gaps, methods — then verify

5. **Manage**

   Zotero: save the 5 keepers with notes ↩ new questions

Overview → core → map → summarize → manage. This compresses a traditionally weeks-long literature review into an afternoon — and step 5's notes feed **new questions**, looping you back to Class 2. It's exactly today's project.

Quick check · tap an answer

You're brand new to a field and don't even know its keywords yet. Which tool is the SMARTEST first move?

- Connected Papers — but you have no seed paper yet
- Zotero — start by organizing (nothing to organize yet)
- A meaning-based tool (Semantic Scholar / Elicit / Perplexity) to surface the vocabulary and seed papers
- Google Scholar with a random keyword guess

CASE STUDY · OUR RUNNING EXAMPLE, MAPPED

Mapping "pedestrian detection for self-driving"

1
Seed

Start from one seminal detector paper — YOLO (you summarized it in Class 4).

2
Branch

Follow forward citations to recent **robustness** work — detection at night, in rain, in crowds.

3
Map

Connected Papers draws how the 5 papers relate; you spot the night-detection cluster.

The whole course, converging: Class 2 asked the night-recall question · Class 3 drew its pipeline · Class 4 summarized YOLO · Class 5 read its detection log · **today you map the literature around it.** Five classes, one thread — and your literature map is where the question meets the field. Prefer another domain? Same three steps.

BEYOND PAPERS · THE COLLABORATION TOOL

GitHub — where research code lives

💾 Version control

Every change to your code saved and reversible — no more `final_v3_REAL_final.py`. Built on Git.

🤝 Collaboration

Many people, one project, no emailed zip files. Issues track tasks; pull requests review changes.

🌟 Your research shopfront

A public repo with a clear README is how researchers **show their work** — reproducible, citable, and a real portfolio.

Why it matters now: from Class 5 on, you're writing real code and notebooks. A GitHub repo is where your project becomes **shareable and permanent** — this course's decks and site live in one. Tonight's homework: create yours.

LIVE DEMO 1

Elicit — a findings table from a question

Type your real research question; watch a comparison table of papers appear. The Class-4 framework, automated across a shelf.

```
You ▸ Does adding low-light training data improve night-time pedestrian detection?
You ▸ Add columns: dataset used · main metric · reported improvement · stated limitation
You ▸ Sort by most relevant; open the top 3 — do their abstracts match Elicit's summary?
You ▸ Which paper's limitation is closest to your own research question?
```

elicit.com

LIVE DEMO 2

Semantic Scholar — meaning + TL;DR + influence

Search by idea, triage by TL;DR, rank by influence. Great when you don't know a field's exact words.

You ▸ Search the CONCEPT: "making object detectors robust to bad weather" (no exact keyword)

You ▸ Read three TL;DRs — which papers survive a 5-second triage?

You ▸ Open one high-influence-score paper → click its "citations" (forward) and "references" (backward)

You ▸ Save the best as your SEED paper for Connected Papers next.

semanticscholar.org

LIVE DEMO 3

Connected Papers — the graph, live

Paste the seed from demo 2; the neighborhood draws itself. Concept 2, on screen.

You ▸ Paste your seed paper → generate the graph

You ▸ Find the CLUSTERS — which knot of papers is your topic's sub-field?

You ▸ Switch to Tree view: which papers are the roots? Which are the frontier?

You ▸ Pick 5 papers total that best surround your question — that's your literature map.

connectedpapers.com

HANDS-ON LAB 1 · READ THE GRAPH IN CODE

Explore a citation graph — roots and frontier

Connected Papers draws the graph; here's what it's **computing**. A tiny citation network of detection papers — find the **roots** (most foundational) and the **frontier** (newest), just like Tree view.

```
# A tiny citation graph: who cites whom in "object detection"
# edge (A, B) means: paper A cites paper B  (A is newer, B is a root)
cites = [
    ("R-CNN", "SelectiveSearch"), ("Fast-RCNN", "R-CNN"),
    ("Faster-RCNN", "Fast-RCNN"), ("YOLO", "R-CNN"),
    ("YOLO", "Faster-RCNN"), ("YOLOv3", "YOLO"),
    ("NightDetect", "YOLOv3"), ("RainRobust", "YOLOv3"),
    ("CrowdPeds", "Faster-RCNN"), ("CrowdPeds", "NightDetect"),
]

papers = set(p for edge in cites for p in edge)
```

▶ Run   🎧 Explain each line   ✨ Explain   🐞 Debug   ⚡ Improve   🤖 Mira   Python loads on first Run

```
▶ Run the code to see the output here.
```

HANDS-ON LAB 2 · RANK PAPERS BY RELEVANCE

A tiny relevance ranker — pick the 5 that matter

Facing 10,000 papers, how do tools rank relevance? One simple idea: **word overlap** between your question and each paper. Rank these candidates against a question — then read the top 5.

```python
# Rank papers by how well they overlap your research question.
question = "improve pedestrian detection accuracy at night with low light"

papers = {
  "A": "Low-light image enhancement for night-time object detection",
  "B": "A survey of pedestrian detection methods in autonomous driving",
  "C": "Fast real-time object detection on daytime highway images",
  "D": "Thermal cameras improve pedestrian detection in darkness",
  "E": "Sentiment analysis of movie reviews using transformers",
  "F": "Night-time low-light pedestrian detection with synthetic data",
}
```

▶ Run | 🎧 Explain each line | ✨ Explain | 🐞 Debug | ⚡ Improve | 🤖 Mira | Python loads on first Run

```
▶ Run the code to see the output here.
```

YOUR PROJECT

Build a literature map

Turn **your** topic into **5 key papers** and a picture of how they relate — the map you'll navigate for the rest of your project.

- Pick your topic (your Class-2 question is perfect)
- Find 5 relevant papers with **Elicit / Semantic Scholar**
- Paste a seed into **Connected Papers** to see the links
- Draw a relationship graph — a one-line note on each paper

Deliverable

A literature map: 5 real papers + a graph showing who builds on whom.

Success looks like

All 5 papers are **real and on-topic** (you opened each), and the graph shows at least one **"builds on"** link.

BUILD IT NOW · HANDS-ON

Run the relay for your own question

1. **Overview:** ask Perplexity "latest advances in <*your topic*>?" — get oriented
2. **Find:** in Elicit or Semantic Scholar, search your question; save 5 promising papers
3. **Read the abstracts** — drop any that aren't truly on-topic (relevance beats quantity)
4. **Map:** paste your best paper into Connected Papers; find the cluster around your question
5. **Draw** the 5-node graph with a one-line note per paper + at least one "builds on" arrow

Elicit | Semantic Scholar | Connected Papers | Perplexity

> **Instructor's rule:** read each abstract **before** adding it. **5 well-chosen papers beat 50 skimmed ones** — and each one you actually read is a citation you can defend.

BENCHMARK · JUDGE AGAINST THIS

What a finished literature map looks like

Topic: *night-time pedestrian detection* — five papers, each with a one-line role:

① **YOLO** (root, 2016) — real-time single-pass detection. *The foundation everyone builds on.*

② **YOLOv3** (2018) — stronger, multi-scale. *builds on ①.*

③ **Low-light enhancement** (2020) — brighten images first. *a different attack on the same problem.*

④ **Thermal-camera detection** (2021) — skip visible light entirely. *the rival approach.*

⑤ **Synthetic night data** (2022) — generate night images to train on. *builds on ①②; closest to my question.*

The gap I found: nobody combines ③ (enhancement) + ⑤ (synthetic data). *That's my study.*

The bar: real papers, a "builds on" arrow (① → ② → ⑤), rival approaches noted (③ vs ④), and a **gap you can point to** — the Swanson move, done with your own map.

REFERENCE · KEY TERMS

Today's vocabulary

Literature review — the systematic map of what's already known about your question.

Citation graph — papers as nodes, "cites" as edges; roots behind, frontier ahead.

Forward / backward citations — who cites this paper (newer) vs. what it cites (foundations).

Semantic search — finding papers by meaning, not exact keywords.

Research gap — the unanswered space between what's known; often between two fields (Swanson).

Reference manager — Zotero/Mendeley: your searchable library of saved papers + notes.

RECOMMENDED HOMEWORK

Before next class

1. **Finish your literature map** — 5 real, on-topic papers + a graph with ≥1 "builds on" link. Submit in Google Classroom. Note which paper is **closest to your own question**.
2. **Create a GitHub research repo** — a `README.md` stating your research direction, goal, and initial plan. Paste the link in 💬. (This is your project's home for the rest of the course.)
3. **Try 2 new tools** from today you'd never used, on your own topic. One sentence each: did it help?

⭐ Stretch (optional)

Find a genuine **Swanson gap**: two sub-fields in your map that almost never cite each other. What question lives in the space between them?

zotero.org — free reference manager

Note: creating a GitHub account requires signing up yourself (do it with a parent if you're under 13). Questions → 💬 on any slide.

TODAY'S LINE TO REMEMBER

"Give me six hours to chop down a tree, and I'll spend the first four sharpening the axe."

— Abraham Lincoln

Today was axe-sharpening. The right tools turn six hours of searching into one.

🔊 Speak this

WRAP-UP

Three things to keep

- Specialized tools **find and connect** papers — Elicit, Semantic Scholar, Connected Papers
- Citation graphs map a whole field: **back to roots, forward to frontier**
- **5 well-chosen papers beat 50 skimmed ones** — read each one you cite

Exit ticket: which of your 5 papers is closest to your own question — and what gap did you spot? Post in 💬.

Next · Class 7 — where data comes from, and how to judge whether it's any good. Stage 3 begins.

BACKUP MATERIAL

Extra slides — if there's time

Deeper dives and extra practice. Skip in a tight hour; pull up on demand or for a longer session.

Tools & creativity

amplifier or crutch?

More of the toolbox

Roboflow, Zotero, Perplexity

Bonus quiz

one more check

BACKUP · EXTENDED THINKING

Do tools kill creativity — or free it?

⚠️ The worry

Over-reliance breeds mental laziness — if the tool always finds the papers, do you lose the ability to think through a field yourself?

✅ The case for

Tools clear away the tedious hours (searching, formatting, transferring) so your energy goes to the **creative** part: the question, the connection, the gap.

The resolution — the course's recurring theme: a tool should be an **amplifier** of your thinking, not a **substitute** for it. Class 1 said it of AI; Class 4 said it of LLMs; today, of your whole toolbox. You steer; the tools row.

BACKUP · THE REST OF THE TOOLBOX

Three more worth knowing

🏷️ Roboflow

Label images for computer-vision projects — bounding boxes, then export straight to YOLO. **This is how the boxes in our AV example get drawn.** (You'll meet it again in the data classes.)

📚 Zotero

Your reference shoebox: one click saves a paper with its citation; organize with tags and notes; auto-formats your bibliography. Free and open-source.

🔮 Perplexity

Class 1's fact-checker, back as a discovery tool: a search engine that answers with cited sources — perfect for the step-1 field overview.

> The toolbox is bigger than four tools — but **don't chase every new one**. Pick a handful, chain them into your workflow, and go do research. The tool is never the point.

BACKUP · BONUS CHECK

Quick check · tap an answer

Elicit hands you a clean findings table summarizing 12 papers. You're on a deadline. What's the responsible move?

- Cite all 12 straight from the table — it's already summarized
- Cite the 5 most relevant from the table without opening them; Elicit is usually right
- Open the 5 you'll actually cite and confirm each summary against the real paper
- Ignore the table; only manual searching is trustworthy